

### 3.3

```
Hello OS Lab
File Opened Successfully
File Closed Successfully

...Program finished with exit code 0
Press ENTER to exit console.█
```

### 3.1

```
CHILD PROCESS
Child PID : 8176
Parent PID : 8172

PARENT PROCESS
Parent PID : 8172
Parent's Parent PID : 8171

...Program finished with exit code 0
Press ENTER to exit console.█
```

3.2

```
Child Process Running
Child Process Completed
Parent Resumes Execution

...Program finished with exit code 0
Press ENTER to exit console.□
```

4.1

```
Enter Number of Processes: 3
Enter Burst Time for P1: 2
Enter Burst Time for P2: 3
Enter Burst Time for P3: 4

Process BT      WT      TAT
P1      2      0      2
P2      3      2      5
P3      4      5      9

Average Waiting Time = 2.33
Average Turnaround Time = 5.33

...Program finished with exit code 0
Press ENTER to exit console.□
```

4.2

```
Enter Number of Processes: 5
Enter Burst Time for P1: 3
Enter Burst Time for P2: 4
Enter Burst Time for P3: 4
Enter Burst Time for P4: 5
Enter Burst Time for P5: 6

Process BT      WT      TAT
P1      3      0      3
P2      4      3      7
P3      4      7      11
P4      5     11     16
P5      6     16     22

Average Waiting Time = 7.40
Average Turnaround Time = 11.80
```

4.3

```
Enter Number of Processes: 4
Enter Burst Time for P1: 3
Enter Priority for P1: 4
Enter Burst Time for P2: 5
Enter Priority for P2: 6
Enter Burst Time for P3: 5
Enter Priority for P3: 7
Enter Burst Time for P4: 5
Enter Priority for P4: 7

Process Priority    BT      WT      TAT
P1      4          3      0      3
P2      6          5      3      8
P3      7          5      8     13
P4      7          5     13     18

Average Waiting Time = 6.00
Average Turnaround Time = 10.50
```

#### 4.4

```
Enter Number of Processes: 2
Enter Burst Time for P1: 3
Enter Burst Time for P2: 4
Enter Time Quantum: 5

Process BT      WT      TAT
P1      3      0      3
P2      4      3      7

Average Waiting Time = 1.50
Average Turnaround Time = 5.00
```

#### 5.1

```
Message received from child: Hello from Child Process

...Program finished with exit code 0
Press ENTER to exit console.█
```